# CS228 : Logic for Computer Science

## Lecture 0: Introduction and logistics

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# Bugs in programs can be costly!

June 4, 1996 - Ariane 5 Flight 501.

"Working code for the Ariane 4 rocket is reused in the Ariane 5, but the Ariane 5's faster engines trigger a bug in an arithmetic routine inside the rocket's flight computer. The error is in the code that converts a 64-bit floating-point number to a 16-bit signed integer. The faster engines cause the 64-bit numbers to be larger in the Ariane 5 than in the Ariane 4, triggering an overflow condition that results in the flight computer crashing."

Source: Simson Garfinkel, Wired Magazine, 2005: `https://www.wired.com/2005/11/historys-worst-software-bugs/`

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June 4, 1996 - Ariane 5 Flight 501.


1993-Intel Pentium floating point divide.

"A silicon error causes Intel's highly promoted Pentium chip to make mistakes when dividing floating-point numbers that occur within a specific range. For example, dividing 4195835.0/3145727.0 yields 1.33374 instead of 1.33382, an error of 0.006 percent... The bug ultimately costs Intel $475 million."

Source: Simson Garfinkel, Wired Magazine, 2005: `https://www.wired.com/2005/11/historys-worst-software-bugs/`

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June 4, 1996 - Ariane 5 Flight 501.


1993-Intel Pentium floating point divide.

2005 - Toyota Prius Bug.

"Toyota announced a recall of 160,000 of its Prius hybrid vehicles following reports of vehicle warning lights illuminating for no reason, and cars' gasoline engines stalling unexpectedly. But unlike the large-scale auto recalls of years past, the root of the Prius issue wasn't a hardware problem - it was a programming error in the smart car's embedded code. The Prius had a software bug."

Source: Simson Garfinkel, Wired Magazine, 2005: `https://www.wired.com/2005/11/historys-worst-software-bugs/`

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How do you catch bugs?

- Software testing not enough always...
- Need mathematically formal and rigorous guarantees.
- How do we reason about software/programs?

# CS228: Logic for Computer Science

## This course in a nutshell

1. How to reason
   - about human thought – isn't that philosophy??
   - about computing objects: systems or machines or programs
2. How to automate this reasoning

## What we need are

1. languages to formalize reasoning: symbolic logic
2. mathematical models of the computing objects: abstractions, transition systems
3. algorithms and tools to automate this whole process!

# What this course is not about and what it is about

## The idea of this course is not to...

- ... cover a specific topic in depth
- ... exhaustively cover all logics, automata, models

## The idea of this course is to...

- ... prepare students to have a background in formal logic.
- ... see where logic occurs in CS and how it can be used for CS.
- ... look at some logics, techniques in detail.
- ...

# Who uses Logic in CS?

How many queries to a logic tool (SAT/SMT solver to be precise) does Amazon Web Services invoke per day?

- Option A: 1 thousand
- Option B: 100 thousand
- Option C: 1 Million
- Option D: 1 Billion

What do they use it for?

**A Billion SMT Queries a Day**
**(Invited Paper)**

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**Abstract.** Amazon Web Services (AWS) is a cloud computing services provider that has made significant investments in applying formal methods to proving correctness of its internal systems and providing assurance of correctness to their end-users. In this paper, we focus on how we built abstractions and eliminated specifications to scale a verification engine for AWS access policies, Zelkova, to be usable by all AWS users. We present milestones from our journey from a thousand SMT invocations daily to an unprecedented billion SMT calls in a span of five years. In this paper, we talk about how the cloud is enabling application of formal methods, key insights into what made this scale of a billion SMT queries daily possible, and present some open scientific challenges for the formal methods community.

**Keywords:** Cloud Computing · Formal Verification · SMT Solving

# Who uses Logic in CS?

## Widely used in industry

- Hardware and software verification
- Cyber-physical systems: Avionics, Automobiles, space!
- AI and databases
- Hot area: explainability in AI/ML!

Including Amazon, Intel, IBM, Microsoft Research, Google, Samsung, TCS, NASA, Mathworks ...

## Widely used in Academia too

- research groups in all major universities around the world
- An interplay of mathematics and computer science, logic and coding, theory and practice.

For a more colorful video, see `https://www.youtube.com/watch?v=AM_gwEKjnGY`

# Course structure

## Topics to be covered in this course:

- Module 1: Symbolic Logics and formal reasoning
- Module 2: The problem of satisfiability and its applications
- Module 3: Hoare Logic and Program verification
- Module 4: Temporal Logics and Model checking

There are advanced courses in each of these topics! CS433, CS615, CS 6104, CS713, CS735, CS738, CS739 But this course binds them all.... :-)


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# Course logistics

## Mode of instruction

- Lectures: 3 hrs per week
- Weekly problem solving & self study: 3+ hrs per week, compulsory!
- Problemsheets will be given every week.
- (Optional) problem solving sessions by TAs: $\sim$ 1 hr per week
- Piazza for doubts and more.
- Office hours, if needed, by appointment.

## Evaluation Disclaimer: Tentative!

- Quizzes (2 or 3) : 30%
- Midsem: 25%
- Final exam: 40%
- Other: Pop quizzes, Class participation : 5%

# Module 1: Logic, a language and calculus for reasoning

1. Men are mortal
2. Socrates is a man

---

Socrates is mortal

Intuitive Rule/Pattern:

1. $\alpha$ are $\beta$
2. $\gamma$ is an $\alpha$

---

$\gamma$ is $\beta$

1. A barber shaves all those who don't shave themselves
2. The barber needs a shave

---

Who shaves the barber?

## A symbolic logic zoo for reasoning

- Propositional logic
- Predicate logic
- Temporal logic
- Modal logic

# Module 2: Satisfiability

## Boolean Propositional Satisfiability

- Is a sentence written in the propositional logic satisfiable?
- That is, is there an assignment that evaluates it to true?
- A central problem in Algorithms, complexity - NP-complete!
- Applications paramount – A whole book of problems that can be reduced to it.
- Our formal reasoning, via logic, will lead to building efficient algorithms that solve many instances of this problem easily!

## Theory behind powerful solvers

- SAT solvers
- SMT solvers

# Module 3: Application to reasoning about programs

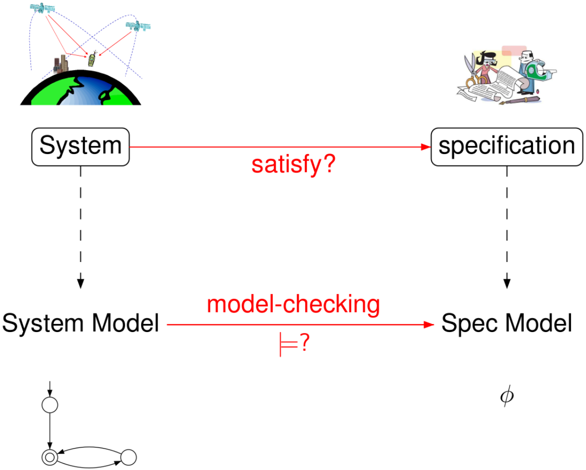
```
int foo(int n) {
    int k, j;
    k = 0; j = 1;
    while (k != n) {
        k = k + 1;
        j = 2*j;
    }
    return(j);
```

```
list * bar(list * i) {
    list *j, *k;
    j = NULL;
    while (i != NULL) {
        k = i→next;
        i→next = j;
        j = i;
        i = k;
    }
    return(i)
```

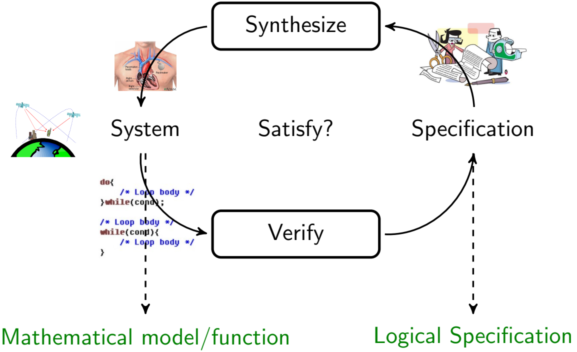
- If "foo" is called with $n > 0$, does it always return $2^n$?
- If "bar" is called with $i$ pointing to an acyclic list, does $i$ always point to an acyclic list when "bar" returns?

- In general, we can't answer this (why?), but then what?
- The "art and science" of proving programs (in)correct – using logic and reasoning!

# Module 4: Application to Model checking



# A Powerful Technique: Formal Verification




- Goal: Develop Efficient Algorithms to Verify and Synthesize
- Challenges: Expressivity of models, Scalability of Algorithms
- We stand on the Shoulders of Giants! Turing Awards in 1996 and 2007 were on this work.

# Overlaps and links to other courses

- Unsurprisingly, there are overlaps with several courses

## Undergrad courses

- Discrete structures (CS105) - reasoning ideas same, same but different!
- Automata theory (CS 310): last part will lead to automata theory

## Other linked courses

- CS 433: Automated reasoning: advanced course more practical
- CS 713: Specialized course on Automata & logic connections
- CS 738: Model checking, cyber-physical systems
- CS 6104: Quantitative Verification
- CS 781: Formal methods and machine learning
- CS 766: Analysis of concurrent programs

# Links to other domains

Programs and Systems are everywhere and everyone wants to make sure they work!

- Machine learning and AI
  - Techniques to verify and explain, check robustness of DNNs.
  - Formal reasoning and logic started AI.
  - Probabilistic logics
  - Prob systems: Markov chains, Markov decision processes, POMDPs
  - Logical expressiveness of GNNs, Transformers, LLMs...
- Programming languages
- Blockchains and other concurrent, distributed algorithms: verifying smart contracts.
- Data bases Specifying properties and using solvers!
- Cryptography security protocols and formalizing attacks!
- Cyber-physical, real-time systems, etc.

# In conclusion

## CS228: Logic for CS

- Formalizing using Logic
- Mathematical Reasoning about programs and systems
- Building efficient algorithms when possible
- A gateway to the fascinating area of formal methods in CS!

- Visit the course webpage for all details!
  `http://www.cse.iitb.ac.in/~akshayss/courses/cs228`